

Unit 29: Area Between Curves — Full Solutions

Warm-up

1. $y = x$ and $y = x^2$ on $[0, 1]$.

Check which is on top: at $x = 0.5$, $x = 0.5$ and $x^2 = 0.25$, so $y = x$ is on top.

$$A = \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \left(\frac{1}{2} - \frac{1}{3} \right) - 0 = \frac{1}{6}$$

Answer: $\frac{1}{6}$.

2. $y = 8 - x^2$ and $y = x^2$.

Intersections: $8 - x^2 = x^2 \Rightarrow 8 = 2x^2 \Rightarrow x = \pm 2$.

At $x = 0$: $8 - x^2 = 8$, $x^2 = 0$, so $8 - x^2$ is on top.

$$\begin{aligned} A &= \int_{-2}^2 [(8 - x^2) - x^2] dx = \int_{-2}^2 (8 - 2x^2) dx = \left[8x - \frac{2x^3}{3} \right]_{-2}^2 \\ &= \left(16 - \frac{16}{3} \right) - \left(-16 + \frac{16}{3} \right) = 16 - \frac{16}{3} + 16 - \frac{16}{3} = 32 - \frac{32}{3} = \frac{64}{3} \end{aligned}$$

Answer: $\frac{64}{3}$.

3. $y = x^2$ and $y = 2x$.

Intersections: $x^2 = 2x \Rightarrow x^2 - 2x = 0 \Rightarrow x(x - 2) = 0 \Rightarrow x = 0, 2$.

At $x = 1$: $x^2 = 1$, $2x = 2$, so $2x$ is on top.

$$A = \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = \left(4 - \frac{8}{3} \right) - 0 = \frac{4}{3}$$

Answer: $\frac{4}{3}$.

4. $y = x + 6$ and $y = x^2$.

Intersections: $x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0 \Rightarrow x = -2, 3$.

At $x = 0$: $x + 6 = 6$, $x^2 = 0$, so $x + 6$ is on top.

$$A = \int_{-2}^3 [(x + 6) - x^2] dx = \left[\frac{x^2}{2} + 6x - \frac{x^3}{3} \right]_{-2}^3$$

At $x = 3$: $\frac{9}{2} + 18 - 9 = \frac{27}{2}$.

At $x = -2$: $2 - 12 + \frac{8}{3} = -10 + \frac{8}{3} = -\frac{22}{3}$.

$$A = \frac{27}{2} - \left(-\frac{22}{3}\right) = \frac{27}{2} + \frac{22}{3} = \frac{81}{6} + \frac{44}{6} = \frac{125}{6}$$

Answer: $\frac{125}{6} \approx 20.83$.

5. $y = 4 - x^2$ and the x-axis.

Intersections: $4 - x^2 = 0 \Rightarrow x = \pm 2$.

$$A = \int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = 16 - \frac{16}{3} = \frac{32}{3}$$

Answer: $\frac{32}{3}$.

6. $y = x$ and $y = x^3$ on $[0, 1]$.

At $x = 0.5$: $x = 0.5$, $x^3 = 0.125$, so $y = x$ is on top.

$$A = \int_0^1 (x - x^3) dx = \left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \left(\frac{1}{2} - \frac{1}{4}\right) - 0 = \frac{1}{4}$$

Answer: $\frac{1}{4}$.

Standard

7. $y = x^2 - 4$ and $y = 4 - x^2$.

Intersections: $x^2 - 4 = 4 - x^2 \Rightarrow 2x^2 = 8 \Rightarrow x = \pm 2$.

At $x = 0$: $x^2 - 4 = -4$, $4 - x^2 = 4$, so $4 - x^2$ is on top — same setup as Problem 2.

$$A = \int_{-2}^2 (8 - 2x^2) dx = \frac{64}{3}$$

Answer: $\frac{64}{3}$.

8. $y = x^3$ and $y = 4x$.

Intersections: $x^3 = 4x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0, \pm 2$.

Check each region: at $x = -1$: $x^3 = -1$, $4x = -4$; since $-1 > -4$, x^3 is on top on $(-2, 0)$.

At $x = 1$: $x^3 = 1$, $4x = 4$; since $4 > 1$, $4x$ is on top on $(0, 2)$.

$$A = \int_{-2}^0 (x^3 - 4x) dx + \int_0^2 (4x - x^3) dx$$

First piece: $\left[\frac{x^4}{4} - 2x^2 \right]_{-2}^0 = 0 - (4 - 8) = 4.$

Second piece: $\left[2x^2 - \frac{x^4}{4} \right]_0^2 = (8 - 4) - 0 = 4.$

$$A = 4 + 4 = 8$$

Answer: 8.

9. $y = \sqrt{x}$ and $y = \frac{x}{2}$ on $[0, 4]$.

Intersections: $\sqrt{x} = \frac{x}{2} \Rightarrow x = \frac{x^2}{4} \Rightarrow 4x = x^2 \Rightarrow x(x - 4) = 0 \Rightarrow x = 0, 4.$

At $x = 1$: $\sqrt{x} = 1$, $\frac{x}{2} = 0.5$, so $\sqrt{x}$ is on top.

$$A = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \left[\frac{2}{3} x^{3/2} - \frac{x^2}{4} \right]_0^4 = \left(\frac{2}{3}(8) - 4 \right) - 0 = \frac{16}{3} - 4 = \frac{4}{3}$$

Answer: $\frac{4}{3}.$

10. $y = 6x(x^2 + 1)^2$, x-axis, $x = 0$, $x = 1$.

Since $x \geq 0$ and $(x^2 + 1)^2 > 0$ throughout $[0, 1]$, the function stays at or above the x-axis — no splitting needed.

Let $u = x^2 + 1$, $du = 2x dx$. When $x = 0$, $u = 1$; when $x = 1$, $u = 2$.

$$\int_0^1 6x(x^2 + 1)^2 dx = 6 \int_1^2 u^2 \cdot \frac{du}{2} = 3 \int_1^2 u^2 du = 3 \left[\frac{u^3}{3} \right]_1^2 = \left[u^3 \right]_1^2 = 8 - 1 = 7$$

Answer: 7.

11. $y = x^2 + 2$ and $y = -x + 4$.

Intersections: $x^2 + 2 = -x + 4 \Rightarrow x^2 + x - 2 = 0 \Rightarrow (x + 2)(x - 1) = 0 \Rightarrow x = -2, 1.$

At $x = 0$: $x^2 + 2 = 2$, $-x + 4 = 4$, so $-x + 4$ is on top.

$$A = \int_{-2}^1 [(-x + 4) - (x^2 + 2)] dx = \int_{-2}^1 (-x^2 - x + 2) dx = \left[-\frac{x^3}{3} - \frac{x^2}{2} + 2x \right]_{-2}^1$$

At $x = 1$: $-\frac{1}{3} - \frac{1}{2} + 2 = \frac{7}{6}.$

At $x = -2$: $\frac{8}{3} - 2 - 4 = -\frac{10}{3}$.

$$A = \frac{7}{6} - \left(-\frac{10}{3}\right) = \frac{7}{6} + \frac{20}{6} = \frac{27}{6} = \frac{9}{2}$$

Answer: $\frac{9}{2} = 4.5$.

12. $y = x^3 - x$ and the x-axis on $[-1, 1]$.

The curve crosses the axis at $x = -1, 0, 1$. On $(-1, 0)$: test $x = -0.5$: $(-0.125) - (-0.5) = 0.375 > 0$, above axis. On $(0, 1)$: test $x = 0.5$: $0.125 - 0.5 = -0.375 < 0$, below axis.

$$A = \int_{-1}^0 (x^3 - x) dx + \int_0^1 [-(x^3 - x)] dx = \int_{-1}^0 (x^3 - x) dx + \int_0^1 (x - x^3) dx$$

First piece: $\left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^0 = 0 - \left(\frac{1}{4} - \frac{1}{2}\right) = 0 - \left(-\frac{1}{4}\right) = \frac{1}{4}$.

Second piece: $\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \left(\frac{1}{2} - \frac{1}{4}\right) - 0 = \frac{1}{4}$.

$$A = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Answer: $\frac{1}{2}$.

Challenge

13. $y = -x^2 + 3x$ and $y = 2x^3 - x^2 - 5x$.

Set equal: $-x^2 + 3x = 2x^3 - x^2 - 5x \Rightarrow 3x = 2x^3 - 5x \Rightarrow 8x - 2x^3 = 0 \Rightarrow 2x(4 - x^2) = 0 \Rightarrow x = 0, \pm 2$.

Let $d(x) = (-x^2 + 3x) - (2x^3 - x^2 - 5x) = -2x^3 + 8x = 2x(4 - x^2)$.

On $(-2, 0)$, test $x = -1$: $2(-1)(3) = -6 < 0$, so $y = 2x^3 - x^2 - 5x$ is on top there.

On $(0, 2)$, test $x = 1$: $2(1)(3) = 6 > 0$, so $y = -x^2 + 3x$ is on top there.

$$A = \int_{-2}^0 [-d(x)] dx + \int_0^2 d(x) dx = \int_{-2}^0 (2x^3 - 8x) dx + \int_0^2 (-2x^3 + 8x) dx$$

First piece: $\left[\frac{x^4}{2} - 4x^2\right]_{-2}^0 = 0 - (8 - 16) = 8$.

Second piece: $\left[-\frac{x^4}{2} + 4x^2\right]_0^2 = (-8 + 16) - 0 = 8$.

$$A = 8 + 8 = 16$$

Answer: 16.

14. $y = x^4 - 4x^2$ and the x-axis.

Roots: $x^2(x^2 - 4) = 0 \Rightarrow x = 0, \pm 2$.

Check the sign: at $x = 1$, $1 - 4 = -3 < 0$; at $x = -1$, $1 - 4 = -3 < 0$. The curve stays at or below the axis throughout $[-2, 2]$ (touching 0 only at the three roots, never crossing back above), so no splitting is needed.

$$A = \int_{-2}^2 [0 - (x^4 - 4x^2)] dx = \int_{-2}^2 (4x^2 - x^4) dx = \left[\frac{4x^3}{3} - \frac{x^5}{5} \right]_{-2}^2$$

$$\text{At } x = 2: \frac{32}{3} - \frac{32}{5} = \frac{160}{15} - \frac{96}{15} = \frac{64}{15}.$$

$$\text{At } x = -2: -\frac{32}{3} + \frac{32}{5} = -\frac{64}{15}.$$

$$A = \frac{64}{15} - \left(-\frac{64}{15} \right) = \frac{128}{15}$$

Answer: $\frac{128}{15}$.

15. $y = x^3 - 3x$ and $y = x$.

Intersections: $x^3 - 3x = x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0, \pm 2$.

Let $d(x) = (x^3 - 3x) - x = x^3 - 4x$.

On $(-2, 0)$, test $x = -1$: $-1 + 4 = 3 > 0$, so $y = x^3 - 3x$ is on top there.

On $(0, 2)$, test $x = 1$: $1 - 4 = -3 < 0$, so $y = x$ is on top there.

$$A = \int_{-2}^0 (x^3 - 4x) dx + \int_0^2 (4x - x^3) dx$$

$$\text{First piece: } \left[\frac{x^4}{4} - 2x^2 \right]_{-2}^0 = 0 - (4 - 8) = 4.$$

$$\text{Second piece: } \left[2x^2 - \frac{x^4}{4} \right]_0^2 = (8 - 4) - 0 = 4.$$

$$A = 4 + 4 = 8$$

Answer: 8.

16. $y = x\sqrt{9 - x^2}$ from $x = 0$ to $x = 3$.

Since $x \geq 0$ and $\sqrt{9 - x^2} \geq 0$ throughout $[0, 3]$, the curve stays at or above the axis — no splitting needed.

Let $u = 9 - x^2$, $du = -2x dx \Rightarrow x dx = -\frac{du}{2}$. When $x = 0$, $u = 9$; when $x = 3$, $u = 0$.

$$\begin{aligned}\int_0^3 x\sqrt{9-x^2} dx &= \int_9^0 \sqrt{u} \cdot \left(-\frac{du}{2}\right) = \frac{1}{2} \int_0^9 \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2}\right]_0^9 \\ &= \frac{1}{3} \left[u^{3/2}\right]_0^9 = \frac{1}{3} (27 - 0) = 9\end{aligned}$$

Answer: 9.

17. $y = 3x^2(x^3 + 1)$ from $x = -1$ to $x = 1$.

Since $x^2 \geq 0$ always, and $x^3 + 1 \geq 0$ for all $x \geq -1$ (because $x^3 \geq -1$ throughout this range), the whole product stays at or above the x-axis on $[-1, 1]$ — no splitting needed.

Let $u = x^3 + 1$, $du = 3x^2 dx$. When $x = -1$, $u = 0$; when $x = 1$, $u = 2$.

$$\int_{-1}^1 3x^2(x^3 + 1) dx = \int_0^2 u du = \left[\frac{u^2}{2}\right]_0^2 = 2 - 0 = 2$$

Answer: 2.

Applied

18. $R_A(t) = 100 + 20t$, $R_B(t) = 80 + 15t$, on $[0, 10]$.

Since $R_A > R_B$ throughout (both the starting value and the growth rate favor A), the extra cumulative revenue is:

$$\int_0^{10} [R_A(t) - R_B(t)] dt = \int_0^{10} (20 + 5t) dt = \left[20t + \frac{5t^2}{2}\right]_0^{10} = (200 + 250) - 0 = 450$$

Answer: Company A earns \$450 more than Company B over the 10 weeks.

19. $r_1(t) = 5 + 2t$, $r_2(t) = 3 + t$, on $[0, 6]$.

$$\int_0^6 [r_1(t) - r_2(t)] dt = \int_0^6 (2 + t) dt = \left[2t + \frac{t^2}{2}\right]_0^6 = (12 + 18) - 0 = 30$$

Answer: Tank 1 has collected 30 more gallons than Tank 2 after 6 minutes.

20. $v_A(t) = 8 + 0.5t$, $v_B(t) = 7 + 0.3t$, on $[0, 4]$.

$$\int_0^4 [v_A(t) - v_B(t)] dt = \int_0^4 (1 + 0.2t) dt = \left[t + 0.1t^2\right]_0^4 = (4 + 1.6) - 0 = 5.6$$

Answer: Runner A gains a lead of 5.6 miles over Runner B.

21. $p_A(x) = 50 + 3x$, $p_B(x) = 40 + 2x$, on $[0, 20]$.

$$\int_0^{20} [p_A(x) - p_B(x)] dx = \int_0^{20} (10 + x) dx = \left[10x + \frac{x^2}{2} \right]_0^{20} = (200 + 200) - 0 = 400$$

Answer: Strategy A generates \$400 more total profit than Strategy B over the 20 weeks.